Analytic solutions of the wobble equation: derivations for §7 and the damped form of §8.1

Working notes (prepared with Claude) — to be checked

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This note derives, step by step, the three results of §7 ("What the equation does") of the note `rotation_dynamic_elastic.pdf`. Section 5 constructs the damped form $\mathsf{L} = \sigma_{\mathrm{e}}\mathsf{E} - \gamma\mathsf{1}$ of N§8.1. Equation numbers prefixed by "N" refer to that note.

1 Setting up

The equation. The starting point is eq. N(22), with no damping ($\gamma = 0$):

$$\dot{\boldsymbol{\omega}} = \sigma_{\mathrm{e}}\,\mathsf{E}\big[(1-\mathsf{K})\boldsymbol{\omega} - \mathbf{q}^{\zeta}(t)\big], \qquad \mathsf{E} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{1}$$

Here $\boldsymbol{\omega}$ is the equatorial rotation vector and $\mathbf{q}^{\zeta}(t)$ the excitation from the direct load.

Isotropic approximation. Take $\mathsf{K} = \kappa\mathsf{1}$. Then $(1-\mathsf{K})\boldsymbol{\omega} - \mathbf{q}^{\zeta} = (1-\kappa)\boldsymbol{\omega} - \mathbf{q}^{\zeta}$. Factoring out $(1-\kappa)$,

$$\dot{\boldsymbol{\omega}} = \sigma_{\mathrm{e}}(1-\kappa)\,\mathsf{E}\Big[\boldsymbol{\omega} - \frac{\mathbf{q}^{\zeta}}{1-\kappa}\Big]. \tag{2}$$

Define the Chandler frequency and the static answer:

$$\sigma_{\mathrm{CW}} := \sigma_{\mathrm{e}}(1-\kappa), \qquad \boldsymbol{\omega}_s(t) := \frac{\mathbf{q}^{\zeta}(t)}{1-\kappa}. \tag{3}$$

$\boldsymbol{\omega}_s$ is the static answer because setting $\dot{\boldsymbol{\omega}} = \mathbf{0}$ in eq. (1) gives $(1-\kappa)\boldsymbol{\omega} = \mathbf{q}^{\zeta}$, since E is invertible. The equation becomes

$$\boxed{\dot{\boldsymbol{\omega}} = \sigma_{\mathrm{CW}}\,\mathsf{E}\,(\boldsymbol{\omega} - \boldsymbol{\omega}_s)} \tag{4}$$

Complex form. Write $m = \omega_1 + i\omega_2$, and similarly m_s for $\boldsymbol{\omega}_s$. For any $\mathbf{v} = (v_1, v_2)$, $\mathsf{E}\mathbf{v} = (-v_2, v_1)$, whose complex form is $-v_2 + iv_1 = i(v_1 + iv_2)$. So

$$\mathsf{E} \;\longleftrightarrow\; \text{multiplication by } i, \qquad \mathsf{E}^T = -\mathsf{E} \;\longleftrightarrow\; \text{multiplication by } -i, \tag{5}$$

and eq. (4) is

$$\boxed{\dot{m} = i\sigma_{\mathrm{CW}}\,(m - m_s)} \tag{6}$$

Multiplying by i rotates by $+90°$, i.e. from $\mathbf{e}_1$ towards $\mathbf{e}_2$. This is anticlockwise viewed from above the north pole, the same sense as the Earth's rotation (*prograde*).

The rotation matrix. Because $\mathsf{E}^2 = -1$, the exponential series splits into even and odd terms:

$$\begin{aligned} e^{\theta\mathsf{E}} &= \sum_{n\ge 0} \frac{\theta^n \mathsf{E}^n}{n!} = \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\Big)\mathsf{1} + \Big(\theta - \frac{\theta^3}{3!} + \dots\Big)\mathsf{E} \\ &= \cos\theta\,\mathsf{1} + \sin\theta\,\mathsf{E} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} =: \mathsf{R}(\theta), \end{aligned} \tag{7}$$

the anticlockwise rotation by θ. In complex form this is $e^{i\theta}$.

2 Constant excitation

Let $\mathbf{q}^\zeta$ be constant, so $\boldsymbol{\omega}_s$ is constant and $\dot{\boldsymbol{\omega}}_s = \mathbf{0}$.

1. Define the departure from the static answer, $\mathbf{e} := \boldsymbol{\omega} - \boldsymbol{\omega}_s$. Then $\dot{\mathbf{e}} = \dot{\boldsymbol{\omega}}$, and eq. (4) gives

$$\dot{\mathbf{e}} = \sigma_{\mathrm{CW}}\, \mathsf{E}\, \mathbf{e}. \tag{8}$$

2. This is a constant-coefficient linear ODE, with solution

$$\mathbf{e}(t) = e^{\sigma_{\mathrm{CW}} \mathsf{E} t}\, \mathbf{e}(0) = \mathsf{R}(\sigma_{\mathrm{CW}} t)\, \mathbf{e}(0), \tag{9}$$

   using eq. (7).

3. Hence

$$\boxed{\boldsymbol{\omega}(t) = \boldsymbol{\omega}_s + \mathsf{R}(\sigma_{\mathrm{CW}} t)\, (\boldsymbol{\omega}_0 - \boldsymbol{\omega}_s)} \tag{10}$$

4. Interpretation:

   - R preserves length, so $|\boldsymbol{\omega} - \boldsymbol{\omega}_s| = |\boldsymbol{\omega}_0 - \boldsymbol{\omega}_s|$ for all t. The pole moves on a circle centred on the static answer.
   - The angle increases at rate σ_{CW}, anticlockwise, so the motion is prograde at the Chandler frequency.
   - Nothing in eq. (10) decays, so the pole never settles.

5. **From rest.** With $\boldsymbol{\omega}_0 = \mathbf{0}$, $\mathbf{e}(0) = -\boldsymbol{\omega}_s$. The radius is $|\boldsymbol{\omega}_s|$, so the free wobble has the same size as the static response, and the circle passes through the origin.

6. **General K.** The same steps with $\mathsf{M} := \sigma_{\mathrm{e}} \mathsf{E}(1 - \mathsf{K})$ and $\boldsymbol{\omega}_s := (1 - \mathsf{K})^{-1} \mathbf{q}^\zeta$ give $\dot{\mathbf{e}} = \mathsf{M}\mathbf{e}$, so $\boldsymbol{\omega}(t) = \boldsymbol{\omega}_s + e^{\mathsf{M}t}(\boldsymbol{\omega}_0 - \boldsymbol{\omega}_s)$. This is the form in N§7. When K is symmetric the eigenvalues of M are $\pm i\sigma_{\mathrm{e}}\sqrt{\det(1 - \mathsf{K})}$ (N§6.3), and the circle becomes a slightly elliptical closed orbit.

3 Periodic excitation

1. **Circular forcing.** Take $m_s(t) = \hat{m}_s\, e^{i\nu t}$ with $\hat{m}_s$ a complex constant (the static answer at the forcing frequency). $\nu > 0$ is prograde circular forcing and $\nu < 0$ retrograde.

2. **Trial solution.** Seek a particular solution at the same frequency, $m(t) = \hat{m}\, e^{i\nu t}$. Then $\dot{m} = i\nu\, \hat{m}\, e^{i\nu t}$.

3. Substitute into eq. (6) and cancel the common factor $e^{i\nu t}$:

$$i\nu\, \hat{m} = i\sigma_{\mathrm{CW}}\, (\hat{m} - \hat{m}_s). \tag{11}$$

4. Cancel i and collect the $\hat{m}$ terms:

$$(\sigma_{\mathrm{CW}} - \nu)\, \hat{m} = \sigma_{\mathrm{CW}}\, \hat{m}_s. \tag{12}$$

5. Divide by $\sigma_{\mathrm{CW}} - \nu$ (assuming $\nu \neq \sigma_{\mathrm{CW}}$), then divide numerator and denominator by σ_{CW}:

$$\boxed{\frac{m_{\mathrm{dynamic}}}{m_{\mathrm{static}}} = \frac{\hat{m}}{\hat{m}_s} = \frac{\sigma_{\mathrm{CW}}}{\sigma_{\mathrm{CW}} - \nu} = \frac{1}{1 - \nu/\sigma_{\mathrm{CW}}}} \tag{13}$$

   This is eq. N(24). The ratio is real, so the dynamic response is exactly in phase or exactly in antiphase with the static one.

6. **Real forcing.** A real vector forcing, e.g. $\boldsymbol{\omega}_s = a\cos(\nu t)\,\mathbf{e}_1$, has complex form $m_s = \frac{a}{2}(e^{i\nu t} + e^{-i\nu t})$. This is a prograde part at $+\nu$ and a retrograde part at $-\nu$. By linearity each is multiplied by its own ratio, with $\nu \to +|\nu|$ and $\nu \to -|\nu|$ respectively. Writing $\nu = 2\pi/T$ and $\sigma_{\mathrm{CW}} = 2\pi/T_{\mathrm{CW}}$:

$$\text{prograde: } \frac{1}{1 - T_{\mathrm{CW}}/T}, \qquad \text{retrograde: } \frac{1}{1 + T_{\mathrm{CW}}/T}. \tag{14}$$

7. **Free wobble.** The full solution adds the homogeneous solution $e^{i\sigma_{\mathrm{CW}} t}c$, a free wobble fixed by the initial condition. The ratio describes the forced response only.

8. **Behaviour.**

   - $T \to \infty$: ratio $\to 1$, the static limit.
   - $T = T_{\mathrm{CW}}$, prograde: the denominator vanishes. This is the resonance, and there is none on the retrograde side because $1 + T_{\mathrm{CW}}/T > 1$.
   - $T < T_{\mathrm{CW}}$, prograde: $T_{\mathrm{CW}}/T > 1$, so the ratio is negative. The response is reversed in sign.
   - $T \ll T_{\mathrm{CW}}$: ratio $\approx -T/T_{\mathrm{CW}}$ prograde and $+T/T_{\mathrm{CW}}$ retrograde, so it is small.

9. **Table values.** For example, with $T_{\mathrm{CW}} = 422.2\,\mathrm{d}$ and $T = 365.25\,\mathrm{d}$, $T_{\mathrm{CW}}/T = 1.156$:

$$\text{prograde } \frac{1}{1 - 1.156} = -6.42, \qquad \text{retrograde } \frac{1}{1 + 1.156} = 0.464.$$

   With $T = 30\,\mathrm{d}$, $T_{\mathrm{CW}}/T = 14.07$, giving -0.077 and 0.066. The other entries of the table in N§7 follow in the same way.

T_{CW}/T	30 d	182.6 d	365.25 d	$\to \infty$
$T_{\mathrm{CW}} = 476.0\,\mathrm{d}$	15.87	2.607	1.303	0
$T_{\mathrm{CW}} = 422.2\,\mathrm{d}$	14.07	2.312	1.156	0

4 Linear trend

Let $\mathbf{q}^\zeta = \dot{\mathbf{q}}^\zeta t$ with $\dot{\mathbf{q}}^\zeta$ constant, starting from rest. Then $\boldsymbol{\omega}_s(t) = \dot{\boldsymbol{\omega}}_s\, t$ with $\dot{\boldsymbol{\omega}}_s = \dot{\mathbf{q}}^\zeta/(1-\kappa)$. In complex form $m_s = b\,t$, with b the complex form of $\dot{\boldsymbol{\omega}}_s$.

1. **Trial solution.** The forcing is linear in t, so try $m_p(t) = b\,t + c$ with c a complex constant. Then $\dot{m}_p = b$ and $m_p - m_s = c$.

2. Substitute into eq. (6):

$$b = i\sigma_{\mathrm{CW}}\, c. \tag{15}$$

3. Solve, using $1/i = -i$:

$$c = \frac{b}{i\sigma_{\mathrm{CW}}} = -\frac{i\,b}{\sigma_{\mathrm{CW}}}. \tag{16}$$

4. Translate back with eq. (5): multiplication by $-i$ is E^T. So

$$\boldsymbol{\omega}_p(t) = \boldsymbol{\omega}_s(t) + \frac{1}{\sigma_{\mathrm{CW}}}\,\mathsf{E}^T\dot{\boldsymbol{\omega}}_s. \tag{17}$$

5. **Initial condition.** Add the homogeneous solution $e^{i\sigma_{\rm CW}t}d$. From rest, $m(0)=0$ requires $m_p(0)+d=c+d=0$, so $d=-c$. Hence

$$\boxed{\boldsymbol{\omega}(t)=\boldsymbol{\omega}_s(t)+\frac{1}{\sigma_{\rm CW}}\mathsf{E}^T\dot{\boldsymbol{\omega}}_s+\boldsymbol{\omega}_{\rm free}(t),\qquad \boldsymbol{\omega}_{\rm free}(t)=-\mathsf{R}(\sigma_{\rm CW}t)\,\frac{1}{\sigma_{\rm CW}}\mathsf{E}^T\dot{\boldsymbol{\omega}}_s} \tag{18}$$

   This is eq. N(25). $|\boldsymbol{\omega}_{\rm free}|=|c|$, so the free wobble has the same amplitude as the offset.

6. **The trend.** The only term growing with t is $\boldsymbol{\omega}_s(t)$, so the long-term rate is exactly the static rate $\dot{\boldsymbol{\omega}}_s$.

7. **Direction of the offset.** $\mathsf{E}^T\dot{\boldsymbol{\omega}}_s$ is $\dot{\boldsymbol{\omega}}_s$ rotated by $-90°$, so it is perpendicular to the trend, on its clockwise side (looking down from the north).

8. **Size of the offset.**

$$\Big|\frac{1}{\sigma_{\rm CW}}\mathsf{E}^T\dot{\boldsymbol{\omega}}_s\Big|=\frac{|\dot{\boldsymbol{\omega}}_s|}{\sigma_{\rm CW}}=|\dot{\boldsymbol{\omega}}_s|\,\frac{T_{\rm CW}}{2\pi}. \tag{19}$$

   This is the distance the static pole travels in time $T_{\rm CW}/2\pi$: 75.8 d for $T_{\rm CW}=476.0$ d and 67.2 d for $T_{\rm CW}=422.2$ d.

9. **Why it is not a lag.** A lag of τ would give $\boldsymbol{\omega}_s(t-\tau)=\boldsymbol{\omega}_s(t)-\tau\dot{\boldsymbol{\omega}}_s$, an offset *parallel* to $\dot{\boldsymbol{\omega}}_s$. The actual offset is perpendicular to it.

10. **General K.** With $\mathsf{M}=\sigma_{\rm e}\mathsf{E}(1-\mathsf{K})$ and $\boldsymbol{\omega}_s=(1-\mathsf{K})^{-1}\mathbf{q}^{\zeta}$, eq. (1) is $\dot{\boldsymbol{\omega}}=\mathsf{M}(\boldsymbol{\omega}-\boldsymbol{\omega}_s)$. The trial solution $\boldsymbol{\omega}_s(t)+\mathbf{c}$ gives $\dot{\boldsymbol{\omega}}_s=\mathsf{M}\mathbf{c}$, so $\mathbf{c}=\mathsf{M}^{-1}\dot{\boldsymbol{\omega}}_s$. For $\mathsf{K}=\kappa 1$, $\mathsf{M}^{-1}=(\sigma_{\rm CW}\mathsf{E})^{-1}=\mathsf{E}^T/\sigma_{\rm CW}$, recovering the result above.

5 The damped form $\mathsf{L}=\sigma_{\rm e}\mathsf{E}-\gamma 1$ (N§8.1)

The elastic theory has no dissipation, so the damping in eq. N(26) is not derived from the elastic equations. It is *constructed*: it is the most general linear law that satisfies four requirements, and it contains one free parameter γ. The construction follows.

Requirements.

(R1) **Linear and first order**, like eq. (1): $\dot{\boldsymbol{\omega}}$ is a linear function of $\boldsymbol{\omega}$ and $\mathbf{q}$.

(R2) **Unchanged static limit.** If $\mathbf{q}$ is constant, $\boldsymbol{\omega}=\mathbf{q}$ must be a steady state, so damping changes how the pole approaches the static answer, not the answer itself.

(R3) **Axisymmetry.** With $I_1=I_2$ and isotropic dissipation, the law cannot depend on the choice of equatorial axes.

(R4) **Elastic limit.** With no dissipation it must reduce to $\dot{\boldsymbol{\omega}}=\sigma_{\rm e}\mathsf{E}(\boldsymbol{\omega}-\mathbf{q})$.

Step 1: form of the law (R1, R2). By (R1), $\dot{\boldsymbol{\omega}}=\mathsf{A}\boldsymbol{\omega}-\mathsf{B}\mathbf{q}$ for constant 2×2 matrices A,B. By (R2), $\boldsymbol{\omega}=\mathbf{q}$ with $\dot{\boldsymbol{\omega}}=\mathbf{0}$ must hold for every constant $\mathbf{q}$, so $(\mathsf{A}-\mathsf{B})\mathbf{q}=\mathbf{0}$ for all $\mathbf{q}$, i.e. $\mathsf{B}=\mathsf{A}$. Writing $\mathsf{L}:=\mathsf{A}$,

$$\dot{\boldsymbol{\omega}}=\mathsf{L}\,(\boldsymbol{\omega}-\mathbf{q}). \tag{20}$$

Step 2: axisymmetry (R3). Rotating the equatorial axes by θ maps $\boldsymbol{\omega} \to \mathsf{R}(\theta)\boldsymbol{\omega}$ and $\mathbf{q} \to \mathsf{R}(\theta)\mathbf{q}$. Eq. (20) keeps its form for every θ only if $\mathsf{R}(\theta)\mathsf{L} = \mathsf{L}\mathsf{R}(\theta)$ for all θ.

1. Differentiate at $\theta = 0$. Since $\mathsf{R}(\theta) = \cos\theta\,\mathsf{1} + \sin\theta\,\mathsf{E}$ from eq. (7), $\mathsf{R}'(0) = \mathsf{E}$. The condition therefore requires $\mathsf{EL} = \mathsf{LE}$. Conversely, if L commutes with E it commutes with every $\cos\theta\,\mathsf{1} + \sin\theta\,\mathsf{E}$.

2. Write $\mathsf{L} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Then

$$\mathsf{LE} = \begin{pmatrix} b & -a \\ d & -c \end{pmatrix}, \qquad \mathsf{EL} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix}. \tag{21}$$

3. Equating entries: $b = -c$ and $a = d$. So

$$\mathsf{L} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} = a\,\mathsf{1} - b\,\mathsf{E}. \tag{22}$$

   Rename the two real constants: $\mathsf{L} = \beta\mathsf{E} - \gamma\mathsf{1}$ with $\beta = -b$ and $\gamma = -a$.

Step 3: elastic limit (R4). With no dissipation, L must equal $\sigma_\mathrm{e}\mathsf{E}$. So $\beta = \sigma_\mathrm{e}$, and γ is the one remaining parameter, which vanishes without dissipation:

$$\boxed{\mathsf{L} = \sigma_\mathrm{e}\mathsf{E} - \gamma\mathsf{1}} \tag{23}$$

Step 4: sign of γ and Q. Take $\mathsf{K} = \kappa\mathsf{1}$, so $\mathbf{q} = \mathbf{q}^\zeta + \kappa\boldsymbol{\omega}$, and set $\mathbf{q}^\zeta = \mathbf{0}$ for the free motion:

1. Eq. (20) gives $\dot{\boldsymbol{\omega}} = \mathsf{L}(1-\kappa)\boldsymbol{\omega} = (1-\kappa)(\sigma_\mathrm{e}\mathsf{E} - \gamma\mathsf{1})\boldsymbol{\omega}$.

2. In complex form ($\mathsf{E} \to i$): $\dot{m} = (1-\kappa)(i\sigma_\mathrm{e} - \gamma)\,m = (i\sigma_\mathrm{CW} - \gamma(1-\kappa))\,m$, using $\sigma_\mathrm{CW} = \sigma_\mathrm{e}(1-\kappa)$.

3. Solution: $m(t) = m_0\,e^{i\sigma_\mathrm{CW}t}\,e^{-\gamma(1-\kappa)t}$. The pole still circles at σ_CW, and its amplitude decays at rate $\gamma(1-\kappa)$. Decay requires $\gamma > 0$.

4. The quality factor of a free oscillation at frequency σ_CW is defined by the amplitude decaying as $e^{-\sigma_\mathrm{CW}t/(2Q)}$. Equating the rates, $\sigma_\mathrm{CW}/(2Q) = \gamma(1-\kappa)$, so

$$Q = \frac{\sigma_\mathrm{CW}}{2\gamma(1-\kappa)} = \frac{\sigma_\mathrm{e}(1-\kappa)}{2\gamma(1-\kappa)} = \frac{\sigma_\mathrm{e}}{2\gamma}, \qquad \gamma = \frac{\sigma_\mathrm{e}}{2Q}. \tag{24}$$

Step 5: relation to Munk & MacDonald. Their damped wobble equation is $\dot{m} = i\tilde{\sigma}\,(m - \chi)$ with a complex frequency $\tilde{\sigma}$. Writing $\tilde{\sigma} = \sigma_\mathrm{e} + i\gamma$ gives $i\tilde{\sigma} = i\sigma_\mathrm{e} - \gamma$, which by eq. (5) is exactly $\sigma_\mathrm{e}\mathsf{E} - \gamma\mathsf{1}$. The construction above is therefore the Munk–MacDonald form, written for the undamped Euler frequency with the feedback K generated by the sea-level iteration rather than folded into $\tilde{\sigma}$.

Remarks.

- Because $\mathbf{q} = \mathbf{q}^\zeta + \mathsf{K}\boldsymbol{\omega}$, $\mathsf{L}(\boldsymbol{\omega} - \mathbf{q}) = \mathsf{L}[(1-\mathsf{K})\boldsymbol{\omega} - \mathbf{q}^\zeta]$. Damping the departure from $\mathbf{q}$ and damping the departure from the static answer are the same thing.

- γ is a single phenomenological constant. Physically the dissipation comes from mantle anelasticity, i.e. frequency-dependent Love numbers (N§9). The constant-γ form reproduces a given Q at the Chandler frequency, but not the frequency dependence away from it.

- With an anisotropic ocean K, (R3) holds only approximately. Assuming isotropic dissipation is then a modelling choice, not a consequence of symmetry.